



Figure 70-1: A soroban

Each column in the soroban represents a different digit. The rightmost column is the ones place, the column to its left is the tens place, the column to the left of that is the hundreds place, and so on. The Q, W, E, R, T, Y, U, I, O, and P keys along the top of your keyboard can increase the digit at their respective positions, while the A, S, D, F, G, H, J, K, L, and ; keys will decrease them. The beads on the virtual soroban will slide to reflect the current number. You can also enter numbers directly.

The four beads below the horizontal divider are “earth” beads, and lifting them up against the divider counts as 1 for that digit. The bead above the horizontal divider is a “heaven” bead, and pulling it down against the divider counts as 5 for that digit, so pulling down one heaven bead and pulling up three earth beads in the tens column represents the number 80. More information about abacuses and how to use them can be found at <https://en.wikipedia.org/wiki/Abacus>.

The Program in Action

When you run *soroban.py*, the output will look like this:

```
Soroban - The Japanese Abacus
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+=====+
I 0 0 0 0 0 0 0 0 0 0 I
I | | | | | | | | | I
I | | | | | | | | | I
+=====+
I | | | | | | | | | I
I | | | | | | | | | I
I 0 0 0 0 0 0 0 0 0 0 I
I 0 0 0 0 0 0 0 0 0 0 I
I 0 0 0 0 0 0 0 0 0 0 I
I 0 0 0 0 0 0 0 0 0 0 I
+==0==0==0==0==0==0==0==0==0==+
+q w e r t y u i o p
-a s d f g h j k l ;
(Enter a number, "quit", or a stream of up/down letters.)
> pppiiii
```